Complex Analysis: Contours, Fourier Transforms, and Schwarz–Christoffel

Problems with background and complete solutions

1 Problem 1: Evaluate $\int_{-1}^{1} \frac{\sqrt{1-x^2}}{x^2+1}\,dx$ by contour integration

Hint. Consider $F(z) = \sqrt{z^2-1}/(z^2+1)$ with a branch cut on $[-1,1]$, and integrate around a large circle.

Background

Fix the branch of $G(z) = \sqrt{z^2-1}$ with cut on $[-1,1]$ and $G(x) > 0$ for $x > 1$. For $x \in (-1,1)$,

$$G(x+i0) = +\,i\sqrt{1-x^2}, \qquad G(x-i0) = -\,i\sqrt{1-x^2}.$$

Hence the sum of the upper and lower lip integrals across the cut equals

$$2i \int_{-1}^{1} \frac{\sqrt{1-x^2}}{x^2+1}\,dx = 2i\,I.$$

Solution

Define

$$F(z) = \frac{\sqrt{z^2-1}}{z^2+1}, \qquad z \in \mathbb{C} \setminus [-1,1].$$

Poles: $z = \pm i$ (simple). Large-circle behavior: $F(z) = \frac{1}{z} + O(z^{-3})$; thus

$$\int_{|z|=R} F(z)\,dz \longrightarrow 2\pi i \qquad (R \to \infty).$$

Deforming the big circle to a contour hugging the cut gives

$$\underbrace{\int_{|z|=R} F(z)\,dz}_{\to 2\pi i} = \underbrace{\int_{\text{upper lip}} F\,dz + \int_{\text{lower lip}} F\,dz}_{=2i\,I} + (\text{vanishing end caps}).$$

Residues at the simple poles:

$$\operatorname{Res}_{z=i} F = \lim_{z\to i} \frac{(z-i)\sqrt{z^2-1}}{(z-i)(z+i)} = \frac{\sqrt{i^2-1}}{2i} = \frac{\sqrt{-2}}{2i} = \frac{\sqrt{2}}{2},$$

$$\operatorname{Res}_{z=-i} F = \lim_{z\to -i} \frac{(z+i)\sqrt{z^2-1}}{(z-i)(z+i)} = \frac{\sqrt{(-i)^2-1}}{-2i} = \frac{\sqrt{-2}}{-2i} = \frac{\sqrt{2}}{2}.$$

Hence, by the residue theorem,

$$\int_{|z|=R} F(z)\,dz \longrightarrow 2\pi i \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \right) = 2\pi i \sqrt{2}.$$

Equating the two expressions for the big-circle integral,

$$2i\,I = 2\pi i(\sqrt{2}-1) \quad \Rightarrow \quad \boxed{I = \pi(\sqrt{2}-1)}.$$

Check (real calculus)

With $x = \sin\theta$,

$$I = 2\int_0^{\pi/2} \frac{\cos^2\theta}{1+\sin^2\theta}\,d\theta = 2\int_0^{\infty} \frac{dt}{(1+t^2)(1+2t^2)}\,dt = \pi(\sqrt{2}-1).$$

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2 Problem 2: Fourier transforms and inverse verification

We use the convention $\widehat{f}(\xi) = \int_{-\infty}^{\infty} f(x)\,e^{-i\xi x}\,dx$ and $f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \widehat{f}(\xi)\,e^{i\xi x}\,d\xi$.

(a) $f(x) = e^{-|x|}$ — Transform and step-by-step inverse

Transform:

$$\widehat{f}(\xi) = \int_0^{\infty} e^{-(1+i\xi)x}\,dx + \int_{-\infty}^{0} e^{(1-i\xi)x}\,dx = \frac{1}{1+i\xi} + \frac{1}{1-i\xi} = \boxed{\frac{2}{1+\xi^2}}.$$

Inverse by residues (two cases):

$$f(x) = \frac{1}{2\pi}\int_{\mathbb{R}} \frac{2\,e^{i\xi x}}{1+\xi^2}\,d\xi = \frac{1}{2\pi}\int_{\mathbb{R}} \frac{2\,e^{i\xi x}}{(\xi-i)(\xi+i)}\,d\xi.$$

Case $x > 0:$ $\quad f(x) = \frac{1}{2\pi}\oint_{\Gamma^+} \frac{2\,e^{i\xi x}}{(\xi-i)(\xi+i)}\,d\xi$ (close upward; Jordan's lemma)

$$= \frac{1}{2\pi}(2\pi i)\ \mathrm{Res}_{\xi=i}\left(\frac{2\,e^{i\xi x}}{(\xi-i)(\xi+i)}\right) = \frac{i\,2\,e^{ix}}{2i(i+i)} = e^{-x}.$$

Case $x < 0:$ $\quad f(x) = \frac{1}{2\pi}\oint_{\Gamma^-} \frac{2\,e^{i\xi x}}{(\xi-i)(\xi+i)}\,d\xi$ (close downward)

$$= \frac{1}{2\pi}(2\pi i)\ \mathrm{Res}_{\xi=-i}\left(\frac{2\,e^{i\xi x}}{(\xi-i)(\xi+i)}\right) = e^{x}.$$

Therefore $\boxed{f(x) = e^{-|x|}}$.

(b) $f(x) = e^{-a^2x^2}$ $(a > 0)$ — Transform and step-by-step inverse

Transform:

$$\widehat{f}(\xi) = \int_{\mathbb{R}} e^{-a^2x^2-i\xi x}\,dx = e^{-\xi^2/(4a^2)}\int_{\mathbb{R}} e^{-a^2\left(x+i\frac{\xi}{2a^2}\right)^2}\,dx$$

$$\Rightarrow \qquad \boxed{\widehat{f}(\xi) = \frac{\sqrt{\pi}}{a}\,e^{-\xi^2/(4a^2)}}.$$

(Justify by integrating over a large rectangle whose vertical sides are the real line and the line shifted up by $i\,\xi/(2a^2)$; the horizontal sides vanish by Gaussian decay.)

Inverse by a rectangular shift (explicit steps):

$$\begin{aligned}
f(x) &= \frac{1}{2\pi}\int_{\mathbb{R}} \frac{\sqrt{\pi}}{a}\, e^{-\xi^2/(4a^2)}\, e^{i\xi x}\, d\xi \\
&= \frac{1}{2\pi}\cdot\frac{\sqrt{\pi}}{a}\int_{\mathbb{R}} \exp\Big(-\frac{1}{4a^2}(\xi-2ia^2x)^2\Big)\, \exp\big(-a^2x^2\big)\, d\xi && \text{(complete the square)} \\
&= \frac{1}{2\pi}\cdot\frac{\sqrt{\pi}}{a}\, e^{-a^2x^2}\int_{\mathbb{R}+2ia^2x} \exp\Big(-\frac{u^2}{4a^2}\Big)\, du && \text{(shift contour vertically)} \\
&= \frac{1}{2\pi}\cdot\frac{\sqrt{\pi}}{a}\, e^{-a^2x^2}\int_{\mathbb{R}} \exp\Big(-\frac{u^2}{4a^2}\Big)\, du && \text{(no poles; horizontal sides vanish)} \\
&= \frac{1}{2\pi}\cdot\frac{\sqrt{\pi}}{a}\, e^{-a^2x^2}\cdot 2a\sqrt{\pi} \;=\; e^{-a^2x^2}.
\end{aligned}$$

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3 Problem 3: Schwarz–Christoffel map from the half-plane to a rectangle

Statement

Write down, as an integral, the Schwarz–Christoffel map from the upper half-plane $\mathbb{H}$ to a rectangle, with vertices being the images of the points $z = \pm 1$ and $z = \pm a$ (with $a > 1$ real). Explain why a cannot be chosen arbitrarily, but is determined by the rectangle's aspect ratio.

Background

For prevertices $x_k \in \mathbb{R}$ mapping to interior angles $\alpha_k \pi$ in the polygon,

$$f'(z) = C\prod_k (z - x_k)^{\alpha_k - 1} \qquad \text{(up to an affine change of variables).}$$

A rectangle has $\alpha_k = \frac{1}{2}$ at each vertex, so every factor's exponent is $-\frac{1}{2}$.

Solution

Place prevertices at $-a, -1, 1, a$ (in order) with $a > 1$. Then

$$f'(z) = C\,(z+a)^{-1/2}(z+1)^{-1/2}(z-1)^{-1/2}(z-a)^{-1/2} = \frac{C}{\sqrt{(z^2-1)(z^2-a^2)}}.$$

Integrating,

$$\boxed{f(z) = A + C\int^{z} \frac{dt}{\sqrt{(t^2-1)(t^2-a^2)}}}$$

maps $\mathbb{H}$ conformally to a rectangle (up to translation and scaling).

Aspect ratio determines a

Side lengths are proportional to real integrals between consecutive prevertices:

$$W(a) = \int_{-1}^{1} \frac{dt}{\sqrt{(1-t^2)(a^2-t^2)}}, \qquad H(a) = \int_{1}^{a} \frac{dt}{\sqrt{(t^2-1)(a^2-t^2)}}.$$

Let $k = 1/a \in (0,1)$. Standard substitutions give

$$W(a) = \frac{2}{a} K(k), \qquad H(a) = \frac{2}{a} K'(k),$$

where K is the complete elliptic integral of the first kind and $K'(k) = K(\sqrt{1-k^2})$. Thus

$$\frac{H(a)}{W(a)} = \frac{K'(k)}{K(k)} \qquad \text{with } k = \frac{1}{a}.$$

Given a rectangle, this ratio fixes k (hence a) uniquely; therefore a is not free.

4(a). Statement and branch points

Suppose $\zeta^2 = z^2 + 1$ and

$$z - i = r_1 e^{i\theta_1}, \qquad z + i = r_2 e^{i\theta_2} \qquad (r_1, r_2 > 0).$$

Then

$$\boxed{\zeta = \pm (r_1 r_2)^{1/2} e^{i(\theta_1+\theta_2)/2}}.$$

Reason. $z^2 + 1 = (z-i)(z+i) = r_1 r_2 e^{i(\theta_1+\theta_2)}$. Any square root has the two signs above. The branch points of the multifunction $(z^2+1)^{1/2}$ are where either factor vanishes, i.e. $z = \pm i$.

4(b). A branch with a finite vertical cut

Define the branch by

$$f(z) = (r_1 r_2)^{1/2} e^{i(\theta_1+\theta_2)/2}, \qquad -\frac{\pi}{2} < \theta_1, \theta_2 \le \frac{3\pi}{2}.$$

Values across the imaginary axis

Let $z = x + iy$ and set $\phi(y) = \frac{1}{2}(\theta_1 + \theta_2)$. Note

$$r_1 = |y-1|, \quad r_2 = |y+1|, \qquad \arg(it) = \begin{cases} \pi/2, & t > 0, \\ -\pi/2, & t < 0. \end{cases}$$

(i) $|y| < 1$. $y - 1 < 0$, $y + 1 > 0$.

$$x \to 0^+ : \theta_1 \to (-\pi/2)^+,\ \theta_2 \to (\pi/2)^- \Rightarrow \phi(y) \to 0,$$
$$x \to 0^- : \theta_1 \to (3\pi/2)^-,\ \theta_2 \to (\pi/2)^+ \Rightarrow \phi(y) \to \pi.$$

Hence

$$f(0^+ + iy) = \sqrt{1-y^2}, \qquad f(0^- + iy) = -\sqrt{1-y^2}.$$

(ii) $y > 1$. $\theta_1 = \theta_2 = \pi/2$ on both sides, so $\phi(y) = \pi/2$ and

$$f(\pm 0 + iy) = i\sqrt{y^2-1}.$$

(iii) $y < -1$. $\theta_1 = \theta_2 = -\pi/2 \pmod{2\pi}$, so $\phi(y) = -\pi/2$ and

$$f(\pm 0 + iy) = -i\sqrt{y^2-1}.$$

Branch cut and behaviour at infinity

The only jump occurs for $|y| < 1$ on the imaginary axis, hence the branch cut is

$$\boxed{S = \{x + iy : x = 0,\ |y| \le 1\}}.$$

For $|z| \to \infty$,

$$f(z) = |z|e^{i\arg z}(1 + o(1)) = z(1 + o(1)),$$

so $f(z) \sim z$. The image of S is the real segment $[-1, 1]$; the imaginary axis outside S maps to the imaginary axis. Thus $f : \mathbb{C} \setminus S \to \mathbb{C} \setminus [-1, 1]$ is conformal with $f(z) \sim z$.

4(c). A branch with two semi-infinite vertical cuts

Now define

$$f(z) = (r_1 r_2)^{1/2} e^{i(\theta_1+\theta_2)/2}, \qquad -\frac{3\pi}{2} < \theta_1 \le \frac{\pi}{2}, \quad -\frac{\pi}{2} < \theta_2 \le \frac{3\pi}{2}.$$

Values across the imaginary axis

Let $z = x + iy$, $x \to 0^{\pm}$.

- $|y| < 1$: we may keep $\theta_1 \approx -\pi/2$, $\theta_2 \approx \pi/2$ on both sides, hence

$$f(\pm 0 + iy) = \sqrt{1-y^2} \quad \text{(no jump)}.$$

- $y > 1$: crossing the axis forces θ_1 to jump by 2π; thus

$$f(0^+ + iy) = +i\sqrt{y^2-1}, \qquad f(0^- + iy) = -i\sqrt{y^2-1}.$$

- $y < -1$: analogously,

$$f(0^+ + iy) = -i\sqrt{y^2-1}, \qquad f(0^- + iy) = +i\sqrt{y^2-1}.$$

Therefore the branch cut consists of the two rays $(-\infty i, -i] \cup [i, +\infty i)$.

Values on the real axis and asymptotics by half-planes

For $z = x \in \mathbb{R}$,

$$r_1 = r_2 = \sqrt{x^2+1}, \qquad \frac{\theta_1+\theta_2}{2} = \begin{cases} 0, & x > 0, \\ \pi, & x < 0. \end{cases}$$

Hence

$$\boxed{f(x) = \operatorname{sgn}(x)\,\sqrt{x^2+1}}.$$

As $|z| \to \infty$,

$$f(z) \sim |z| e^{i \arg z} \times \begin{cases} 1, & \Re z > 0, \\ -1, & \Re z < 0, \end{cases}$$

i.e.

$$\boxed{f(z) \sim z \text{ for } \Re z > 0, \qquad f(z) \sim -z \text{ for } \Re z < 0.}$$

Exact problem statement (as in the text)

4. Branches of $\sqrt{z^2+1}$ — Problem statement

(a) **(a)** Show that, if $\zeta^2 = z^2 + 1$ and

$$z = i + r_1 e^{i\theta_1} = -\,i + r_2 e^{i\theta_2}, \qquad r_1, r_2 > 0,\ \theta_1, \theta_2 \in \mathbb{R},$$

then

$$\zeta \;=\; \pm\,(r_1 r_2)^{1/2}\, e^{\,i(\theta_1+\theta_2)/2}.$$

Explain briefly why $z = \pm i$ are the branch points of the multifunction $(z^2+1)^{1/2}$.

(b) **(b)** Consider the branch defined by

$$f(z) \;=\; (r_1 r_2)^{1/2}\, e^{\,i(\theta_1+\theta_2)/2}, \qquad -\tfrac{\pi}{2} < \theta_1,\ \theta_2 \le \tfrac{3\pi}{2}.$$

State the values of $(\theta_1+\theta_2)/2$ on either side of the imaginary axis, and hence compute $f(\pm 0 + iy)$ in terms of y for $y \in \mathbb{R}$. Deduce that the branch cut is

$$S \;=\; \{\, x + iy :\ x = 0,\ |y| \le 1 \,\}.$$

Show that $f(z) \sim z$ as $|z| \to \infty$ (i.e. $f(z)/z \to 1$ as $|z| \to \infty$). Sketch the image of the cut z-plane $\mathbb{C} \setminus S$ under the map $\zeta = f(z)$.

(c) **(c)** Consider instead the branch

$$f(z) \;=\; (r_1 r_2)^{1/2}\, e^{\,i(\theta_1+\theta_2)/2}, \qquad -\tfrac{3\pi}{2} < \theta_1 \le \tfrac{\pi}{2}, \quad -\tfrac{\pi}{2} < \theta_2 \le \tfrac{3\pi}{2}.$$

State the values of $(\theta_1+\theta_2)/2$ on either side of the imaginary axis, and hence compute $f(\pm 0 + iy)$ in terms of y for $y \in \mathbb{R}$. Deduce that the branch cut is along the imaginary axis from $z = -i\infty$ to $z = -i$ and from $z = i$ to $z = i\infty$. Compute $f(x)$ in terms of x for $x \in \mathbb{R}$. Show that

$$f(z) \sim \begin{cases} z, & \text{as } |z| \to \infty \text{ with } \Re(z) > 0, \\ -z, & \text{as } |z| \to \infty \text{ with } \Re(z) < 0. \end{cases}$$

Sketch the images of the half-planes $\Re(z) > 0$ and $\Re(z) < 0$ under the map $\zeta = f(z)$.

Solutions (with background)

Throughout, write

$$z - i = r_1 e^{i\theta_1}, \qquad z + i = r_2 e^{i\theta_2}, \qquad r_1, r_2 > 0,$$

so that

$$z^2 + 1 = (z - i)(z + i) = r_1 r_2\, e^{i(\theta_1+\theta_2)}.$$

(a) Formula for ζ and branch points

Since $\zeta^2 = r_1 r_2\, e^{i(\theta_1+\theta_2)}$, any square root is

$$\boxed{\zeta = \pm (r_1 r_2)^{1/2} e^{i(\theta_1+\theta_2)/2}}.$$

Branch points occur at zeros of $z^2 + 1$, where an argument for a factor cannot be chosen continuously: these are exactly $z = \pm i$.

(b) Branch with finite vertical cut $S = \{x = 0,\ |y| \le 1\}$

Branch choice. Take $-\frac{\pi}{2} < \theta_1, \theta_2 \le \frac{3\pi}{2}$ and define $f(z) = (r_1 r_2)^{1/2} e^{i(\theta_1+\theta_2)/2}$.

Angles on the imaginary axis. Let $z = x + iy$ and denote $\phi(y) = \frac{1}{2}(\theta_1 + \theta_2)$. Use

$$r_1 = |y - 1|, \quad r_2 = |y + 1|, \qquad \arg(it) = \begin{cases} \pi/2, & t > 0, \\ -\pi/2, & t < 0. \end{cases}$$

(i) $|y| < 1$. As $x \to 0^+$: $\theta_1 \to (-\pi/2)^+$, $\theta_2 \to (\pi/2)^- \Rightarrow \phi(y) \to 0$. As $x \to 0^-$: to keep θ_1 in $(-\pi/2, 3\pi/2]$ we add 2π, so $\theta_1 \to (3\pi/2)^-$, $\theta_2 \to (\pi/2)^+ \Rightarrow \phi(y) \to \pi$. Hence

$$\boxed{f(0^+ + iy) = \sqrt{1 - y^2}, \qquad f(0^- + iy) = -\sqrt{1 - y^2} \quad (|y| < 1)}.$$

(ii) $y > 1$. $\theta_1 = \theta_2 = \pi/2$ from either side, so

$$\boxed{f(\pm 0 + iy) = i\sqrt{y^2 - 1} \quad (y > 1)}.$$

(iii) $y < -1$. $\theta_1 = \theta_2 = -\pi/2 \pmod{2\pi}$, so

$$\boxed{f(\pm 0 + iy) = -i\sqrt{y^2 - 1} \quad (y < -1)}.$$

Branch cut. The only sign jump occurs for $|y| < 1$ on the imaginary axis, hence

$$\boxed{S = \{x + iy :\ x = 0,\ |y| \le 1\}}.$$

Asymptotics. As $|z| \to \infty$,

$$r_1 r_2 = |z - i||z + i| = |z^2 + 1| = |z|^2(1 + o(1)), \qquad \theta_1 + \theta_2 = \arg(z^2 + 1) = 2\arg z + o(1),$$

so

$$\boxed{f(z) \sim z \quad (|z| \to \infty)}.$$

Image sketch. The cut segment S maps to the real interval $[-1, 1]$ (opposite sides map to $(0, 1]$ and $[-1, 0)$); outside S the imaginary axis maps to the imaginary axis. Thus $f : \mathbb{C}\backslash S \to \mathbb{C}\backslash[-1, 1]$ is conformal with $f(z) \sim z$.

(c) Branch with two semi-infinite cuts

Branch choice. Take $-\frac{3\pi}{2} < \theta_1 \le \frac{\pi}{2}$ and $-\frac{\pi}{2} < \theta_2 \le \frac{3\pi}{2}$, and define $f(z) = (r_1 r_2)^{1/2} e^{i(\theta_1+\theta_2)/2}$.

Angles across the imaginary axis. For $z = x + iy$:

(i) $|y| < 1$. We may keep $\theta_1 \approx -\pi/2$ and $\theta_2 \approx \pi/2$ on both sides, hence

$$\boxed{f(\pm 0 + iy) = \sqrt{1-y^2} \quad (|y|<1)}.$$

(ii) $y > 1$. Crossing the axis forces θ_1 to jump by 2π; therefore

$$\boxed{f(0^+ + iy) = +i\sqrt{y^2-1}, \qquad f(0^- + iy) = -i\sqrt{y^2-1} \quad (y>1)}.$$

(iii) $y < -1$. Similarly,

$$\boxed{f(0^+ + iy) = -i\sqrt{y^2-1}, \qquad f(0^- + iy) = +i\sqrt{y^2-1} \quad (y<-1)}.$$

Branch cuts. The discontinuities occur on the rays

$$\boxed{(-i\infty, -i] \,\cup\, [i, +i\infty)}.$$

Values on the real axis. For $z = x \in \mathbb{R}$, $r_1 = r_2 = \sqrt{x^2+1}$ and

$$\frac{\theta_1+\theta_2}{2} = \begin{cases} 0, & x>0, \\ \pi, & x<0. \end{cases} \qquad \Rightarrow \qquad \boxed{f(x) = \operatorname{sgn}(x)\sqrt{x^2+1}}.$$

Asymptotics by half-planes. As $|z| \to \infty$,

$$\boxed{f(z) \sim z \ \text{ for } \Re z > 0, \qquad f(z) \sim -z \ \text{ for } \Re z < 0}.$$

4 Problem 5: Images under $\zeta = \dfrac{1}{z}$, $\zeta = \sin z$, and $\zeta = z + e^z$

(a) Intersection of $|z-1| < 1$ and $|z+i| < 1$ under $\zeta = 1/z$

Background (inversion). If $w = 1/z$ and $|z-a| = |a|$ (a circle through 0), then

$$|1 - aw| = |a||w| \iff 1 = 2\Re(aw) \iff \Re(aw) = \tfrac{1}{2},$$

i.e. the image is a straight line.

Boundary images.

$$|z-1| = 1 \longmapsto \Re(w) = \tfrac{1}{2}, \qquad |z+i| = 1 \longmapsto \Re((-i)w) = \tfrac{1}{2} \iff \Im(w) = \tfrac{1}{2}.$$

Which side is the interior?

$$|z-1| < 1 \iff |1/w - 1| < 1 \iff |1-w| < |w| \iff \Re(w) > \tfrac{1}{2};$$

$$|z+i|<1 \iff |1+iw|<|w| \iff \Im(w)>\tfrac{1}{2}.$$

Image. The lens $|z-1|<1 \cap |z+i|<1$ maps to the quarter-plane

$$\boxed{\{\, w:\ \Re w>\tfrac{1}{2},\ \Im w>\tfrac{1}{2}\,\}}.$$

(b) Strip $-\dfrac{\pi}{2}<x<\dfrac{\pi}{2}$ under $\zeta=\sin z$

Write $z=x+iy$. Then

$$\sin z=\sin x\,\cosh y\ +\ i\ \cos x\ \sinh y.$$

Boundary lines.

$$\sin\Big(\pm\frac{\pi}{2}+iy\Big)=\pm\cosh y\in\mathbb{R},\quad |\cosh y|\ge 1.$$

Hence the boundaries map onto the real rays $[1,\infty)$ and $(-\infty,-1]$.

Conformality. $\cos z\neq 0$ on $|x|<\pi/2$ (zeros lie on $x=\pi/2+k\pi$), so sin is conformal and injective on the strip.

Image. The image is the simply connected domain whose boundary is those rays; by symmetry (the line $x=0$ maps to the entire imaginary axis),

$$\boxed{\sin:\ \{-\tfrac{\pi}{2}<\Re z<\tfrac{\pi}{2}\}\ \xrightarrow[1\text{–}1]{}\ \mathbb{C}\setminus\big((-\infty,-1]\cup[1,\infty)\big).}$$

(c) Strip $-\pi<y<\pi$ under $\zeta=f(z)=z+e^z$

Let $z=x+iy$. Then

$$f(z)=\big(x+e^x\cos y\big)\ +\ i\big(y+e^x\sin y\big).$$

Critical points (hint).

$$f'(z)=1+e^z=0 \iff e^z=-1 \iff z=i(2k+1)\pi.$$

There are no critical points in the open strip $-\pi<y<\pi$; the boundary points $z=\pm i\pi$ are the only adjacent critical points.

Boundary images. For $y=\pm\pi$, $\sin y=0$, $\cos y=-1$:

$$f(x\pm i\pi)=\big(x-e^x\big)\ \pm\ i\pi.$$

Here $u(x)=x-e^x$ has a unique maximum -1 at $x=0$ and $u(x)\to-\infty$ as $x\to\pm\infty$. Thus each boundary maps to a horizontal half-line

$$(-\infty,-1]\,+\,i\pi,\qquad (-\infty,-1]\,-\,i\pi,$$

meeting at the critical values $-1\pm i\pi$.

Interior and image. With no interior critical points, f is conformal on the strip. For fixed $y \in (-\pi, \pi)$, $f(x + iy) \to -\infty + iy$ as $x \to -\infty$ and $f(x + iy) \sim e^x e^{iy} \to \infty$ along the ray $\arg \zeta = y$ as $x \to +\infty$. Hence the image is the plane with the two horizontal half-lines removed:

$$\boxed{\mathbb{C} \setminus \Big((-\infty, -1] + i\pi \ \cup \ (-\infty, -1] - i\pi \Big).}$$

5 Problem 6 — Conformal mappings

Exact problem statement

(a) D is the region exterior to the two circles $|z - 1| = 1$ and $|z + 1| = 1$. Find a conformal mapping from D to the exterior of the unit circle.
Hint: First apply an inversion with respect to the origin ($z \mapsto 1/z$), then rotate and scale so as to use the exponential function to get to a half plane, from where use Möbius.

(b) Find a conformal map from the unit disc $|z| < 1$ onto the strip $-\pi/2 < \eta < \pi/2$, taking the origin to the origin, 1 to $\xi = +\infty$, -1 to $\xi = -\infty$. (Thus, the upper half of the unit circle maps to $\eta = \pi/2$ and the lower half to $\eta = -\pi/2$.)

(c) Find a conformal map of the quarter-disc $0 < |z| < 1$, $0 < \arg z < \pi/2$ to the upper half-plane $\eta > 0$, taking 0 to 0, 1 to 1, and i to ∞.

Solutions (step by step)

(a) Exterior of the two circles $\to$ exterior of $|\zeta| = 1$.

- **Invert.** Let $w = 1/z$. Then the circles through 0 become lines:

  $$|z - 1| = 1 \Rightarrow \Re w = \tfrac{1}{2}, \qquad |z + 1| = 1 \Rightarrow \Re w = -\tfrac{1}{2}.$$

  Hence D maps to the vertical strip $S = \{-\frac{1}{2} < \Re w < \frac{1}{2}\}$.
- **Rotate/scale to a horizontal strip.** Put $s = i\pi w$. Then $|\Im s| < \pi/2$.
- **Exponential to a half-plane.** Let $u = e^s$. Then $|\Im s| < \pi/2 \implies \Re u > 0$.
- **Möbius to exterior unit disk.** Use

  $$\Phi(u) = \frac{u + 1}{u - 1}, \qquad \operatorname{Re} u > 0 \longmapsto |\zeta| > 1.$$

Therefore a concrete map is

$$\boxed{\zeta = \Phi\big(e^{i\pi/z}\big) = \frac{e^{i\pi/z} + 1}{e^{i\pi/z} - 1}}.$$

(b) Unit disk $\to$ strip $-\pi/2 < \Im\zeta < \pi/2$.

- **Disk to right half-plane (Cayley).**

$$w = \frac{1+z}{1-z}, \qquad |z| < 1 \mapsto \Re w > 0,$$

  and $z = 0 \mapsto w = 1$, $z = 1 \mapsto w = \infty$, $z = -1 \mapsto w = 0$.

- **Log to a horizontal strip.**

$$\zeta = \log w \quad \text{(principal branch)}, \qquad \Im\zeta \in \left(-\tfrac{\pi}{2}, \tfrac{\pi}{2}\right).$$

  Then $\zeta(0) = 0$, $\zeta(1) = +\infty$, $\zeta(-1) = -\infty$.

Hence

$$\boxed{\zeta = \log\left(\frac{1+z}{1-z}\right)}.$$

(c) Quarter-disk $\to$ upper half-plane; $0 \mapsto 0$, $1 \mapsto 1$, $i \mapsto \infty$.

- **Square to upper semicircle.**

$$t = z^2 : \quad \{0 < |z| < 1,\ 0 < \arg z < \tfrac{\pi}{2}\} \longmapsto \{|t| < 1,\ \Im t > 0\}.$$

  Moreover $0 \mapsto 0$, $1 \mapsto 1$, $i \mapsto -1$.

- **Real Möbius to upper half-plane.** Find $M(t)$ with $M(-1) = \infty$, $M(0) = 0$, $M(1) = 1$. A suitable choice is

$$M(t) = \frac{2t}{t+1}.$$

Therefore

$$\boxed{\zeta = M(z^2) = \frac{2z^2}{z^2+1}}.$$

It maps the quarter-disk conformally onto $\Im\zeta > 0$ and has $\zeta(0) = 0$, $\zeta(1) = 1$, $\zeta(i) = \infty$.

6(c) Quarter-disk $0 < |z| < 1,\ 0 < \arg z < \frac{\pi}{2} \longrightarrow$ upper half-plane $\Im\zeta > 0$, with $0 \mapsto 0$, $1 \mapsto 1$, $i \mapsto \infty$

Goal

Construct an explicit conformal map $\zeta = \zeta(z)$ that maps the open quarter-disk

$$Q = \{\, z : \ 0 < |z| < 1,\ 0 < \arg z < \tfrac{\pi}{2} \,\}$$

onto the upper half-plane $\mathbb{H} = \{\zeta : \ \Im\zeta > 0\}$, sending the three marked boundary points

$$z = 0 \longmapsto \zeta = 0, \qquad z = 1 \longmapsto \zeta = 1, \qquad z = i \longmapsto \zeta = \infty.$$

Step 1: Square to unfold the angle

Let

$$t = z^2.$$

Write $z = re^{i\theta}$ with $0 < r < 1$ and $0 < \theta < \frac{\pi}{2}$. Then

$$t = r^2 e^{i(2\theta)} \quad \Rightarrow \quad |t| < 1, \;\; 0 < \arg t < \pi \iff \Im t > 0.$$

Therefore the squaring map $z \mapsto t = z^2$ sends Q *bijectively and conformally* onto the *open upper semicircle*

$$S = \{\, t : \; |t| < 1, \; \Im t > 0 \,\}.$$

On the boundary rays/arcs we have

$$0 \mapsto 0, \qquad 1 \mapsto 1, \qquad i \mapsto -1.$$

(Conformality is immediate, since $(z^2)' = 2z \neq 0$ on Q.)

Step 2: Real Möbius map fixing the three marked points and sending S to $\mathbb{H}$

Seek a real-coefficient Möbius transformation

$$M(t) = \frac{at + b}{ct + d}, \qquad a, b, c, d \in \mathbb{R}, \;\; ad - bc \neq 0,$$

such that

$$M(-1) = \infty, \qquad M(0) = 0, \qquad M(1) = 1.$$

Imposing the conditions:

- $M(-1) = \infty \;\Rightarrow\; c(-1) + d = 0 \;\Rightarrow\; d = c.$
- $M(0) = 0 \;\Rightarrow\; b/d = 0 \;\Rightarrow\; b = 0.$
- $M(1) = 1 \;\Rightarrow\; a/(c + d) = 1 \;\Rightarrow\; a = c + d = 2c.$

Up to an overall nonzero constant (which does not change the map), we can take $c = 1$ and obtain

$$\boxed{M(t) = \frac{2t}{t + 1}}.$$

Why M maps S into $\Im\zeta > 0$ (and in fact maps the whole upper half-plane to itself). Let $t = x + iy$ with $y > 0$. Then

$$M(t) = \frac{2t}{1 + t}, \qquad \Im M(t) = \frac{2\,\Im\big(t(1 + \bar{t})\big)}{|1 + t|^2} = \frac{2\,\Im t}{|1 + t|^2} = \frac{2y}{|1 + t|^2} > 0.$$

Hence M maps $\{\, \Im t > 0 \,\}$ into itself; in particular it maps the upper semicircle S (which lies in $\Im t > 0$) into the upper half-plane. Moreover

$$M'(t) = \frac{2}{(1 + t)^2} \neq 0 \quad \text{on } S,$$

so M is conformal (locally injective) there. The three-point normalization guarantees that it sends the three distinguished boundary points as prescribed:

$$M(-1) = \infty, \qquad M(0) = 0, \qquad M(1) = 1.$$

Step 3: Compose and verify

Define the desired map

$$\boxed{\zeta(z) = M\big(z^2\big) = \frac{2z^2}{z^2+1}}.$$

Then the normalization follows immediately:

$$\zeta(0) = 0, \qquad \zeta(1) = 1, \qquad \zeta(i) = \infty.$$

Because $t = z^2$ has $\Im t > 0$ on Q and $\Im M(t) > 0$ for $\Im t > 0$, we have

$$\Im \zeta(z) = \Im\big(M(z^2)\big) > 0 \quad \text{for all } z \in Q.$$

Also $\zeta'(z) = M'(z^2) \cdot 2z = \dfrac{4z}{(1+z^2)^2} \neq 0$ on Q, hence the map is conformal throughout the domain.

Boundary correspondence (for intuition)

- The *real radius* segment $\{\, 0 < z < 1 \,\}$ maps by $t = z^2$ to $(0,1)$ and then by M to $(0,1) \subset \mathbb{R}$; thus that edge lands on the real axis between 0 and 1.
- The *imaginary radius* segment $\{\, 0 < iz < i \,\}$ maps by $t = z^2$ to $(0,-1)$ and then by M to $(0,\infty)$ on $\mathbb{R}$; thus the edge $0 \to i$ lands on the ray $[0,\infty)$.
- The *circular arc* $\{\, e^{i\theta} : 0 < \theta < \frac{\pi}{2} \,\}$ maps by $t = z^2$ to the unit semicircle $\{\, e^{i\phi} : 0 < \phi < \pi \,\}$ and then by M to the real line as a boundary set.

The interior of the quarter-disk maps onto $\Im\zeta > 0$ by the positivity computation for $\Im M(t)$.

Conclusion

The composition

$$\boxed{\zeta(z) = \frac{2z^2}{z^2+1}}$$

is a conformal bijection from the open quarter-disk Q onto the upper half-plane $\mathbb{H}$, satisfying the three mapping conditions $0 \mapsto 0$, $1 \mapsto 1$, $i \mapsto \infty$.

(b) Unit disk $\to$ strip $-\dfrac{\pi}{2} < \Im\zeta < \dfrac{\pi}{2}$

Disk to right half-plane (Cayley).

$$w = \frac{1+z}{1-z}, \qquad |z| < 1 \implies \operatorname{Re} w > 0,$$

and on the marked points

$$z = 0 \mapsto w = 1, \qquad z = 1 \mapsto w = \infty, \qquad z = -1 \mapsto w = 0.$$

Moreover, the unit circle $|z| = 1 \setminus \{1\}$ maps to the imaginary axis $\operatorname{Re} w = 0$.

Log to a horizontal strip. Use the principal logarithm on the right half-plane:

$$\zeta = \log w = \ln|w| + i\,\arg(w), \qquad \arg(w) \in \left(-\tfrac{\pi}{2}, \tfrac{\pi}{2}\right),$$

so

$$-\frac{\pi}{2} < \Im\zeta < \frac{\pi}{2}.$$

At the marked points:

$$\zeta(0) = \log 1 = 0, \qquad \zeta(1) = \log(\infty) = +\infty, \qquad \zeta(-1) = \log(0) = -\infty.$$

Which boundary goes where. For $z = e^{i\theta}$,

$$w = \frac{1+e^{i\theta}}{1-e^{i\theta}} = -\,i\,\cot\!\left(\frac{\theta}{2}\right) \in i\mathbb{R}.$$

Hence

$$0 < \theta < \pi \text{ (upper semicircle)} \Rightarrow \Im w < 0 \Rightarrow \Im\zeta = -\frac{\pi}{2},$$

$$\pi < \theta < 2\pi \text{ (lower semicircle)} \Rightarrow \Im w > 0 \Rightarrow \Im\zeta = +\frac{\pi}{2}.$$

(If one wants the upper semicircle to land on $\Im\zeta = +\pi/2$, use $\zeta = -\log\!\left(\frac{1+z}{1-z}\right)$.)

Hence

$$\boxed{\zeta = \log\!\left(\frac{1+z}{1-z}\right)}$$

is a conformal bijection from $|z| < 1$ onto the strip $-\pi/2 < \Im\zeta < \pi/2$ with $0 \mapsto 0$, $1 \mapsto +\infty$, $-1 \mapsto -\infty$.

(a) Exterior of the two circles $|z-1| = 1$ and $|z+1| = 1$ $\longrightarrow$ exterior of $|\zeta| = 1$

Goal. Let

$$D = \big\{z \in \mathbb{C} : \ |z-1| \geq 1,\ |z+1| \geq 1\big\} \setminus \big(\{|z-1| = 1\} \cup \{|z+1| = 1\}\big),$$

the region *exterior to both* unit circles centered at ± 1. Construct an explicit conformal bijection $F : D \to \{\zeta : \ |\zeta| > 1\}$.

Step 1: Inversion. Let $w = \dfrac{1}{z}$. For any $a \in \mathbb{C}$, the circle $\{|z-a| = |a|\}$ (which passes through 0) satisfies

$$|z-a| = |a| \iff \left|\frac{1}{w} - a\right| = |a| \iff |1-aw| = |a||w| \iff 1 = 2\,\Re(aw).$$

Hence circles through 0 invert to *lines.* For $a = 1$ and $a = -1$ we obtain

$$|z-1| = 1 \iff \Re w = \frac{1}{2}, \qquad |z+1| = 1 \iff \Re w = -\frac{1}{2}.$$

Which side is the exterior? Using $|1-aw|^2 = (1 - 2\Re(aw) + |a|^2|w|^2)$ we compute

$$|z-1| > 1 \iff |1-w| > |w| \iff 1 - 2\Re w > 0 \iff \Re w < \frac{1}{2},$$

$$|z+1| > 1 \iff |1+w| > |w| \iff 1 + 2\Re w > 0 \iff \Re w > -\frac{1}{2}.$$

Therefore the inversion $w = 1/z$ sends D bijectively onto the *vertical strip*

$$S = \Big\{ w \in \mathbb{C} : \ -\tfrac{1}{2} < \Re w < \tfrac{1}{2} \Big\}.$$

(Since $z \neq 0$ on D, the inversion is analytic and injective there.)

Step 2: Rotate/scale to a horizontal strip. Define

$$s = i\pi w.$$

Write $w = u + iv$ with $-\frac{1}{2} < u < \frac{1}{2}$. Then $s = i\pi(u+iv) = -\pi v + i\pi u$, so

$$|\Im s| = |\pi u| < \frac{\pi}{2}.$$

Consequently S is mapped conformally onto the horizontal strip

$$T = \Big\{ s \in \mathbb{C} : \ |\Im s| < \frac{\pi}{2} \Big\}.$$

Step 3: Exponential to the right half-plane. Set

$$u = e^{\,s}.$$

If $s = \sigma + i\tau$ with $|\tau| < \pi/2$, then

$$\Re u = \Re\big(e^{\sigma}(\cos\tau + i\sin\tau)\big) = e^{\sigma}\cos\tau > 0,$$

because $\cos\tau > 0$ on $(-\pi/2, \pi/2)$. Hence $e^{(\cdot)}$ maps T conformally and bijectively onto the *right half-plane*

$$H = \{\, u \in \mathbb{C} : \ \Re u > 0 \,\},$$

and the boundary lines $\Im s = \pm\pi/2$ go to the imaginary axis $\Re u = 0$.

Step 4: Möbius map from the right half-plane to the exterior disk. Consider

$$\Phi(u) = \frac{u+1}{u-1}.$$

For $u \in H$ we have the equivalence

$$|\Phi(u)| > 1 \iff |u+1| > |u-1| \iff (u+1)(\bar{u}+1) > (u-1)(\bar{u}-1) \iff 4\,\Re u > 0,$$

which holds precisely on H. Thus Φ maps H conformally onto $\{\zeta : \ |\zeta| > 1\}$ and sends the imaginary axis to the unit circle $|\zeta| = 1$.

Step 5: Composition and explicit formula. The composition of the four conformal bijections yields

$$F : D \xrightarrow{z\mapsto w=1/z} S \xrightarrow{w\mapsto s=i\pi w} T \xrightarrow{s\mapsto u=e^{s}} H \xrightarrow{u\mapsto\Phi(u)} \{|\zeta|>1\}.$$

Therefore an explicit mapping is

$$\boxed{\zeta = F(z) = \Phi\big(e^{i\pi/z}\big) = \frac{e^{i\pi/z}+1}{e^{i\pi/z}-1}}.$$

Every factor is analytic and injective on the relevant domain; hence F is conformal and bijective from D onto $\{|\zeta|>1\}$.

Asymptotic check at infinity (sanity). As $z\to\infty$ we have $e^{i\pi/z} = 1+\dfrac{i\pi}{z}+O(z^{-2})$, so

$$\zeta = \frac{2+\dfrac{i\pi}{z}+O(z^{-2})}{\dfrac{i\pi}{z}+O(z^{-2})} = \frac{2}{i\pi}\,z\Big(1+O(z^{-1})\Big) = -\frac{2i}{\pi}\,z\Big(1+O(z^{-1})\Big),$$

which confirms that far out in D the map behaves like an affine scaling of z, as expected.

Boundary correspondence (orientation).

- The line $\Re w = \frac{1}{2}$ (image of $|z-1|=1$) becomes $\Im s = \frac{\pi}{2}$, then the imaginary axis, and finally the unit circle $|\zeta|=1$.
- The line $\Re w = -\frac{1}{2}$ (image of $|z+1|=1$) becomes $\Im s = -\frac{\pi}{2}$, then the same imaginary axis, and again $|\zeta|=1$.
- The interior $-\frac{1}{2}<\Re w<\frac{1}{2}$ maps to $|\Im s|<\frac{\pi}{2}$, then to $\Re u>0$, and finally to $|\zeta|>1$.

This completes the construction and detailed verification.

6 Problem 7: Temperature in a right half–plane with a removed disk

Exact statement (verbatim)

The domain D consists of the right-hand half plane $x>0$ with the circle $|z-a|=b$, $0<b<a$, and its interior removed. Find the temperature $u(x,y)$ in steady heat flow if $u=0$ on the y axis, $u=1$ on $|z-a|=b$, and $u\to 0$ at infinity.

Hint: Show that the mapping $\zeta = \dfrac{z-\alpha}{z+\alpha}$, with α real and positive, takes D onto an annular region with the imaginary axis mapping to $|\zeta|=1$ and show that, if $\alpha^2=a^2-b^2$, then the image of D is a concentric circular annulus.

Background: why the suggested Möbius map produces an annulus

Consider the real-parameter Möbius map

$$\zeta = \zeta(z) = \frac{z-\alpha}{z+\alpha}, \qquad \alpha > 0.$$

Imaginary axis. For $z = iy$ we have

$$\left|\frac{iy-\alpha}{iy+\alpha}\right| = 1 \quad \text{(numerator and denominator are conjugates)},$$

so the y–axis maps to the unit circle $|\zeta| = 1$.

Image of the circle $|z-a| = b$. Write $z = a + be^{i\theta}$ and compute

$$|\zeta|^2 = \frac{|a-\alpha+be^{i\theta}|^2}{|a+\alpha+be^{i\theta}|^2} = \frac{A+B\cos\theta}{C+D\cos\theta},$$

where

$$A = (a-\alpha)^2 + b^2, \qquad B = 2b(a-\alpha), \qquad C = (a+\alpha)^2 + b^2, \qquad D = 2b(a+\alpha).$$

For $|\zeta|$ to be constant on the circle (i.e. the image is a circle *centered at the origin*), the $\cos\theta$–dependence must drop out:

$$\frac{A+B\cos\theta}{C+D\cos\theta} \equiv \frac{A}{C} \quad \Longleftrightarrow \quad AD = BC.$$

We now verify that

$$AD - BC = 2b\Big[(a+\alpha)((a-\alpha)^2+b^2) - (a-\alpha)((a+\alpha)^2+b^2)\Big] = 4b\alpha\,(\alpha^2+b^2-a^2).$$

Thus $AD = BC$ is equivalent to

$$\boxed{\alpha^2 = a^2 - b^2}.$$

With this choice (note $a > b > 0 \Rightarrow \alpha > 0$), the ratio becomes *constant*:

$$|\zeta|^2 = \frac{A}{C} = \frac{(a-\alpha)^2+b^2}{(a+\alpha)^2+b^2} = \frac{2a(a-\alpha)}{2a(a+\alpha)} = \frac{a-\alpha}{a+\alpha}.$$

Hence the circle $|z-a| = b$ maps to the circle $|\zeta| = \rho$ with

$$\boxed{\rho = \sqrt{\frac{a-\alpha}{a+\alpha}} \in (0,1).}$$

Image of the domain. Therefore, for $\alpha = \sqrt{a^2-b^2}$, the map $z \mapsto \zeta$ sends the domain D (right half–plane with the closed disk $\{|z-a| \le b\}$ removed) onto the concentric circular *annulus*

$$\mathcal{A} = \{\ \rho < |\zeta| < 1\ \}.$$

The boundary pieces correspond as follows:

$$\{x = 0\} \longmapsto |\zeta| = 1, \qquad \{|z-a| = b\} \longmapsto |\zeta| = \rho.$$

Solving the Dirichlet problem on the annulus

Let $U(\zeta)$ denote the temperature in the ζ–plane. Harmonicity is preserved by conformal maps, so U is harmonic on $\mathcal{A}$ and the boundary data become

$$U = 0 \quad \text{on} \quad |\zeta| = 1, \qquad U = 1 \quad \text{on} \quad |\zeta| = \rho.$$

By rotational symmetry, U depends only on $r = |\zeta|$. The general radial harmonic function on an annulus is $A \log r + B$. Imposing the boundary values gives

$$U(r) = \frac{\log(1/r)}{\log(1/\rho)} = \frac{-\log r}{-\log \rho} = \frac{\log r}{\log \rho} \quad \text{for} \quad \rho < r < 1.$$

(Any of the three displayed forms is correct; we will use the first one momentarily.)

Pullback to the original z–plane and final formula

The physical temperature is $u(z) = U(\zeta(z))$ with $r = |\zeta(z)| = \left|\frac{z-\alpha}{z+\alpha}\right|$. Hence

$$u(z) = U\left(\left|\frac{z-\alpha}{z+\alpha}\right|\right) = \frac{\log\left(\frac{1}{\left|\frac{z-\alpha}{z+\alpha}\right|}\right)}{\log\left(\frac{1}{\rho}\right)} = \frac{\log\left|\frac{z+\alpha}{z-\alpha}\right|}{\log\left(\frac{1}{\rho}\right)}.$$

Since $\rho = \sqrt{\dfrac{a-\alpha}{a+\alpha}}$ we have

$$\log\left(\frac{1}{\rho}\right) = \frac{1}{2}\log\left(\frac{a+\alpha}{a-\alpha}\right).$$

Therefore a convenient explicit form is

$$\boxed{u(z) = \frac{2\log\left|\dfrac{z+\alpha}{z-\alpha}\right|}{\log\left(\dfrac{a+\alpha}{a-\alpha}\right)}, \qquad \alpha = \sqrt{a^2-b^2}.}$$

Checks of the boundary conditions and far field

- **On the y–axis $x = 0$:** for $z = iy$ we have $\left|\dfrac{z-\alpha}{z+\alpha}\right| = 1$, so the numerator $\log\left|\dfrac{z+\alpha}{z-\alpha}\right| = 0$ and hence $u = 0$.
- **On the circle $|z-a| = b$:** by construction $|\zeta| = \rho$, i.e. $\left|\dfrac{z-\alpha}{z+\alpha}\right| = \rho$, so $\log\left|\dfrac{z+\alpha}{z-\alpha}\right| = \log\left(\dfrac{1}{\rho}\right)$ and the fraction evaluates to $u = 1$.
- **At infinity:** as $|z| \to \infty$, $\left|\dfrac{z+\alpha}{z-\alpha}\right| \to 1$, so the numerator $\to 0$ and $u(z) \to 0$.

Summary

With $\alpha = \sqrt{a^2 - b^2}$ and $\rho = \sqrt{\dfrac{a-\alpha}{a+\alpha}}$, the Möbius map $\zeta = \dfrac{z-\alpha}{z+\alpha}$ carries D onto the annulus $\{\rho < |\zeta| < 1\}$. The unique harmonic function with $U = 1$ on $|\zeta| = \rho$ and $U = 0$ on $|\zeta| = 1$ is $U(r) = \log(1/r)/\log(1/\rho)$, and hence

$$u(z) = \frac{2\,\log\left|\dfrac{z+\alpha}{z-\alpha}\right|}{\log\left(\dfrac{a+\alpha}{a-\alpha}\right)}.$$

This u satisfies all the prescribed boundary conditions and $u(z) \to 0$ as $|z| \to \infty$.

Hyperbolic maps: sinh, cosh, tanh, coth — background, zeros/poles, and strip mappings

1. $w = \sinh z$

Background: entire, odd; period $2\pi i$. **Zeros:** $z = i\pi k$; **poles:** none.
Decomposition: $w = u + iv = \sinh x \cos y + i\, \cosh x \sin y$.
Vertical lines $x = \text{const}$: $\left(\frac{u}{\sinh x}\right)^2 + \left(\frac{v}{\cosh x}\right)^2 = 1$ (ellipses, foci $\pm i$).
Horizontal lines $y = \text{const}$: $\left(\frac{u}{\cos y}\right)^2 - \left(\frac{v}{\sin y}\right)^2 = -1$ (hyperbolas, foci $\pm i$).
Strip mapping:

$$\boxed{\sinh :\ \{|\Im z| < \tfrac{\pi}{2}\} \xrightarrow{1\text{–}1} \mathbb{C} \setminus i\big((-\infty, -1] \cup [1, \infty)\big)}$$

Boundaries $y = \pm\frac{\pi}{2}$ map to imaginary rays $\pm i[1, \infty)$.

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2. $w = \cosh z$

Background: entire, even; period $2\pi i$. **Zeros:** $z = i\pi(k + \frac{1}{2})$; **poles:** none.
Decomposition: $w = u + iv = \cosh x \cos y + i\, \sinh x \sin y$.
Vertical lines $x = \text{const}$: $\left(\frac{u}{\cosh x}\right)^2 + \left(\frac{v}{\sinh x}\right)^2 = 1$ (ellipses, foci ± 1).
Horizontal lines $y = \text{const}$: $\left(\frac{u}{\cos y}\right)^2 - \left(\frac{v}{\sin y}\right)^2 = 1$ (hyperbolas, foci ± 1).
Strip mapping:

$$\boxed{\cosh :\ \{|\Im z| < \pi\} \xrightarrow{2\text{–}1} \mathbb{C} \setminus \big((-\infty, -1] \cup [1, \infty)\big)}$$

On the half-strip $0 < |\Im z| < \pi$, the map is 1–1 onto the slit plane.

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3. $w = \tanh z$

Background: meromorphic, odd; period $i\pi$. **Zeros:** $z = i\pi k$; **poles:** $z = i\pi(k + \frac{1}{2})$.
Strip → disk:

$$\boxed{\tanh : \ \{|\Im z| < \tfrac{\pi}{2}\} \xrightarrow{1\text{--}1} \mathbb{D}}$$

Real line maps to $(-1, 1)$; as $|\Im z| \to \pi/2$, $|\tanh z| \to 1$.

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4. $w = \coth z$

Background: meromorphic, odd; period $i\pi$. **Zeros:** $z = i\pi(k + \frac{1}{2})$; **poles:** $z = i\pi k$.
Strip → exterior disk:

$$\boxed{\coth : \ \{|\Im z| < \tfrac{\pi}{2}\} \setminus \{0\} \xrightarrow{1\text{--}1} \{\, |w| > 1 \,\}}$$

Real line maps to $(-\infty, -1) \cup (1, \infty)$; $|\Im z| \to \pi/2$ gives $|w| \to 1^+$.

Hyperbolic Conformal Maps: sinh, cosh, tanh, coth

Notation and preliminaries

Throughout, write $z = x + iy$, $x, y \in \mathbb{R}$, and $w = u + iv$ for the image. Recall the basic identities

$$\cosh z = \frac{e^z + e^{-z}}{2}, \qquad \sinh z = \frac{e^z - e^{-z}}{2}, \qquad \tanh z = \frac{\sinh z}{\cosh z}, \qquad \coth z = \frac{\cosh z}{\sinh z}.$$

We use the principal branch of the logarithm log on $\mathbb{C} \setminus (-\infty, 0]$, and principal square roots $\sqrt{\cdot}$ with branch cut on $(-\infty, 0]$ unless stated otherwise.

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1. $w = \sinh z$

Analytic background.
sinh is an *entire*, *odd* function, with imaginary-periodicity:

$$\sinh(-z) = -\sinh z, \qquad \sinh(z + 2\pi i) = \sinh z.$$

Zeros occur at $z_k = i\pi k$ for $k \in \mathbb{Z}$ (all simple). No poles.

Cauchy–Riemann decomposition.
With $z = x + iy$,

$$\sinh z = \sinh x \cos y \ + \ i \ \cosh x \sin y \quad \Rightarrow \quad u = \sinh x \cos y, \quad v = \cosh x \sin y.$$

From this, one obtains orthogonal families of confocal conics:

$$\begin{aligned} x = \text{const}: &\quad \left(\frac{u}{\sinh x}\right)^2 + \left(\frac{v}{\cosh x}\right)^2 = 1 &\quad &\text{(ellipses, foci at } \pm i), \\ y = \text{const}: &\quad \left(\frac{u}{\cos y}\right)^2 - \left(\frac{v}{\sin y}\right)^2 = -1 &\quad &\text{(hyperbolas, foci at } \pm i). \end{aligned}$$

Real and imaginary axes map as:

$$y = 0: \ w = \sinh x \in \mathbb{R} \ (\text{onto } \mathbb{R}), \qquad x = 0: \ w = i \sin y \ (\text{onto the segment } i[-1, 1]).$$

Jacobian and conformality.
$\sinh'(z) = \cosh z$, and $\cosh z \neq 0$ on the strip $|\Im z| < \frac{\pi}{2}$; hence sinh is conformal there.

Canonical strip mapping (global picture).

$$\boxed{\sinh: \ \{\, |\Im z| < \tfrac{\pi}{2} \,\} \xrightarrow[\text{biholomorphism}]{} \mathbb{C} \setminus i\big((-\infty, -1] \cup [1, \infty)\big) \ .}$$

Explanation. The boundary lines $y = \pm\frac{\pi}{2}$ map to the imaginary rays $\pm i[1, \infty)$ (via $v = \cosh x \, \sin y$ with $\sin(\pm\pi/2) = \pm 1$ and $u = \sinh x \cos(\pm\pi/2) = 0$), while the interior maps injectively since $\cosh z \neq 0$ and periodicity in the imaginary direction has period 2π, not π.

Inverse branch and branch cuts.
A principal inverse on the slit domain is

$$\operatorname{arsinh} w = \log\big(w + \sqrt{w^2 + 1}\big),$$

with branch cuts along $i(-\infty, -1] \cup i[1, \infty)$ so that $\Im(\operatorname{arsinh} w) \in (-\frac{\pi}{2}, \frac{\pi}{2})$.

Useful identities.

$$\cosh^2 z - \sinh^2 z = 1, \qquad \sinh(z + \zeta) = \sinh z \, \cosh \zeta + \cosh z \, \sinh \zeta.$$

---

2. $w = \cosh z$

Analytic background.
cosh is *entire*, *even*, and $2\pi i$-periodic:

$$\cosh(-z) = \cosh z, \qquad \cosh(z + 2\pi i) = \cosh z.$$

Zeros at $z = i\pi\big(k + \frac{1}{2}\big)$, all simple. No poles.

Cauchy–Riemann decomposition.
With $z = x + iy$,

$$\cosh z = \cosh x \cos y \; + \; i \, \sinh x \sin y \quad \Rightarrow \quad u = \cosh x \cos y, \quad v = \sinh x \sin y.$$

Again one gets confocal conics:

$$x = \text{const} : \quad \left(\frac{u}{\cosh x}\right)^2 + \left(\frac{v}{\sinh x}\right)^2 = 1 \qquad \text{(ellipses, foci at } \pm 1\text{)},$$

$$y = \text{const} : \quad \left(\frac{u}{\cos y}\right)^2 - \left(\frac{v}{\sin y}\right)^2 = 1 \qquad \text{(hyperbolas, foci at } \pm 1\text{)}.$$

On the axes,

$$y = 0 : \; w = \cosh x \in [1, \infty), \qquad x = 0 : \; w = \cos y \in [-1, 1].$$

Critical points and injectivity on strips.
$\cosh'(z) = \sinh z$, which vanishes at $z = i\pi k$. Thus cosh fails to be injective across lines $y = k\pi$, but is injective on any strip avoiding these lines (e.g. $0 < \Im z < \pi$).

Canonical strip mapping.

$$\boxed{\cosh : \; \{0 < \Im z < \pi\} \xrightarrow[\text{biholomorphism}]{} \mathbb{C} \setminus \big((-\infty, -1] \cup [1, \infty)\big) \; .}$$

Explanation. The midline $y = 0$ maps to $[1, \infty)$; the top boundary $y = \pi$ maps to $(-\infty, -1]$; injectivity holds on $0 < \Im z < \pi$ since there $\sinh z \neq 0$.

Inverse branch and branch cuts.

$$\operatorname{arcosh} w = \log\Big(w + \sqrt{w-1}\,\sqrt{w+1}\Big),$$

with branch cuts along $(-\infty, -1] \cup [1, \infty)$ so that $\Re(\operatorname{arcosh} w) \geq 0$ and $0 < \Im(\operatorname{arcosh} w) < \pi$ on the principal sheet.

Useful identities.

$$\cosh(z + \zeta) = \cosh z \, \cosh \zeta + \sinh z \, \sinh \zeta, \qquad \cosh z = 1 + 2 \sinh^2 \frac{z}{2}.$$

---

3. $w = \tanh z$

Analytic background.
tanh is *meromorphic*, *odd*, and $i\pi$-periodic:

$$\tanh(-z) = -\tanh z, \qquad \tanh(z + i\pi) = \tanh z.$$

Zeros at $z = i\pi k$; poles (simple) at $z = i\pi\big(k + \frac{1}{2}\big)$.

Strip $\leftrightarrow$ disk (Cayley-type).

$$\boxed{\tanh : \{ |\Im z| < \tfrac{\pi}{2} \} \xrightarrow[\text{biholomorphism}]{} \mathbb{D}\ .}$$

Boundary correspondence. The real axis $y = 0$ maps bijectively onto $(-1, 1)$ (strictly increasing). As $y \to \pm\frac{\pi}{2}$, $|\tanh(x + iy)| \to 1$ uniformly in x; hence the horizontal boundaries map to $\partial\mathbb{D}$.

Inverse (disk $\to$ strip).

$$\operatorname{artanh} w = \frac{1}{2} \log\left(\frac{1+w}{1-w}\right), \qquad |w| < 1,$$

which maps $\mathbb{D}$ conformally onto $\{|\Im z| < \frac{\pi}{2}\}$.

Derivative and Schwarz–Pick.

$$(\tanh z)' = \operatorname{sech}^2 z, \qquad \left|\tanh'(z)\right| = \frac{1}{|\cosh z|^2}$$

and conjugating by a disk automorphism shows tanh is extremal for the disk-strip version of Schwarz–Pick.

Level sets.
Lines $x = \text{const}$ and $y = \text{const}$ in the strip map to orthogonal circle arcs inside $\mathbb{D}$ meeting $\partial\mathbb{D}$ orthogonally (images of preimages under the conformal equivalence with the half-plane followed by a Cayley map).

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4. $w = \coth z$

Analytic background.
coth is *meromorphic*, *odd*, and $i\pi$-periodic:

$$\coth(-z) = -\coth z, \qquad \coth(z + i\pi) = \coth z.$$

Poles (simple) at $z = i\pi k$; zeros (simple) at $z = i\pi\big(k + \frac{1}{2}\big)$.

Strip $\leftrightarrow$ exterior disk.

$$\boxed{\coth : \{ |\Im z| < \tfrac{\pi}{2} \} \setminus \{0\} \xrightarrow[\text{biholomorphism}]{} \{ |w| > 1 \}\ .}$$

Boundary correspondence. The real axis (excluding 0) maps to the rays $(-\infty, -1) \cup (1, \infty)$. As $y \to \pm\frac{\pi}{2}$, $|\coth(x + iy)| \to 1^+$; thus horizontal boundaries map to the unit circle from the *exterior* side.

Inverse (exterior disk → strip).

$$\operatorname{arcoth} w = \frac{1}{2}\log\left(\frac{w+1}{w-1}\right), \qquad |w| > 1,$$

with the standard logarithm branch so that $\Im(\operatorname{arcoth} w) \in (-\frac{\pi}{2}, \frac{\pi}{2})$.

Relation with $\tanh$**.**

$$\coth z = \tanh\bigl(z + \tfrac{i\pi}{2}\bigr),$$

so the exterior-disk picture is just the disk picture shifted vertically by $i\pi/2$ and inverted via $w \mapsto 1/w$.

---

5. Global relations, inverse formulas, and composition patterns

Inverse formulas (principal branches).

$$\operatorname{arsinh} w = \log\bigl(w + \sqrt{w^2+1}\bigr), \qquad \operatorname{arcosh} w = \log\bigl(w + \sqrt{w-1}\,\sqrt{w+1}\bigr),$$
$$\operatorname{artanh} w = \frac{1}{2}\log\left(\frac{1+w}{1-w}\right), \qquad \operatorname{arcoth} w = \frac{1}{2}\log\left(\frac{w+1}{w-1}\right).$$

Branch cuts are chosen to match the canonical target domains described above.

Disk/half-plane equivalences via Cayley.
Using the Cayley map $C(\zeta) = \frac{1+\zeta}{1-\zeta}$ and its inverse $C^{-1}(w) = \frac{w-1}{w+1}$,

$$\tanh z = \ C^{-1}\bigl(e^{2z}\bigr), \qquad \coth z = \ C^{-1}\bigl(e^{2z}\bigr)^{-1}, \qquad \sinh z = \frac{e^z - e^{-z}}{2}, \quad \cosh z = \frac{e^z + e^{-z}}{2}.$$

Thus strip ↔ disk/exterior-disk mappings for tanh and coth are just the exponential's strip-to-sector equivalences composed with a Cayley map.

Growth and boundary behavior.
As $x \to \pm\infty$ with y fixed, $\sinh z \sim \frac{1}{2}e^x e^{iy}$ and $\cosh z \sim \frac{1}{2}e^x e^{iy}$; hence horizontal lines map to asymptotically radial curves. For tanh and coth, $|\tanh(x+iy)| \to 1$ and $|\coth(x+iy)| \to 1$ exponentially in $|x|$ (uniform in y within the strip height).

Critical sets and injectivity regions.
$\sinh'(z) = \cosh z$ vanishes along $y = \frac{\pi}{2} + \pi\mathbb{Z}$; $\cosh'(z) = \sinh z$ vanishes along $y = \pi\mathbb{Z}$. Therefore, the natural injectivity strips are $|\Im z| < \frac{\pi}{2}$ for sinh and $0 < \Im z < \pi$ for cosh. For tanh and coth, poles on the horizontal boundaries determine the natural strip height π.

---

6. Summary: canonical conformal equivalences (boxed)

$$\boxed{\sinh:\ \{\,|\Im z|<\tfrac{\pi}{2}\,\}\ \longrightarrow\ \mathbb{C}\setminus i\big((-\infty,-1]\cup[1,\infty)\big)\ \ \text{(biholomorphism)}\ .}$$

$$\boxed{\cosh:\ \{\,0<\Im z<\pi\,\}\ \longrightarrow\ \mathbb{C}\setminus\big((-\infty,-1]\cup[1,\infty)\big)\ \ \text{(biholomorphism)}\ .}$$

$$\boxed{\tanh:\ \{\,|\Im z|<\tfrac{\pi}{2}\,\}\ \longrightarrow\ \mathbb{D}\ \ \text{(biholomorphism)}\ .}$$

$$\boxed{\coth:\ \{\,|\Im z|<\tfrac{\pi}{2}\,\}\setminus\{0\}\ \longrightarrow\ \{\,|w|>1\,\}\ \ \text{(biholomorphism)}\ .}$$

7. (Optional) Proof sketches for strip mappings

Injectivity for $\sinh$ **on** $|\Im z|<\pi/2$. If $\sinh z_1=\sinh z_2$ then $e^{z_1}-e^{-z_1}=e^{z_2}-e^{-z_2}$. Rearranging gives $\{e^{z_1},e^{-z_1}\}=\{e^{z_2},e^{-z_2}\}$. Taking logs in the horizontal strip $|\Im z|<\pi/2$ (where the exponential is injective modulo $2\pi i$) forces $z_1=z_2$.

Boundary images for $\sinh$. Set $z=x\pm i\pi/2$. Then $\sinh z=\sinh x\cdot 0\ \pm i\,\cosh x$, so the boundaries map to $\pm i[1,\infty)$; the interior cannot reach those rays by the open mapping theorem and the maximum modulus principle applied to $1/(\sinh z\mp i)$.

Analogous arguments apply to $\cosh$ (use $\sinh$ zeros to avoid critical lines) and to $\tanh,\coth$ using the representation via exponentials and the Cayley transform.

Inverse Circular Functions: arcsin, arccos, arctan, arccot

Conventions

We use principal branches of $\log$ and $\sqrt{\cdot}$ with branch cut $(-\infty,0]$ unless stated otherwise. For each inverse, the principal domain is obtained by slitting the w–plane along the images of the multiple values.

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1. $z=\arcsin w$

Principal formula (single-valued on the slit plane).

$$\boxed{\arcsin w=-\,i\,\log\Big(i\,w+\sqrt{1-w^2}\Big)}$$

Branch points & cuts.
Branch points at $w=\pm1$; principal cuts along $(-\infty,-1]\cup[1,\infty)$.

Target strip (canonical range).

$$\boxed{\arcsin:\ \mathbb{C}\setminus\big((-\infty,-1]\cup[1,\infty)\big)\ \longrightarrow\ \{\,|\Re z|<\tfrac{\pi}{2}\,\}}$$

For $w\in(-1,1)$, $\arcsin w\in(-\frac{\pi}{2},\frac{\pi}{2})\subset\mathbb{R}$. For $w>1$, $\arcsin w=\frac{\pi}{2}-i\ \operatorname{arcosh} w$ (hence $\Re=\frac{\pi}{2}$); for $w<-1$, $\arcsin w=-\frac{\pi}{2}+i\ \operatorname{arcosh}(-w)$ (hence $\Re=-\frac{\pi}{2}$).

Boundary correspondence.
Approaching the cuts $[1,\infty)$ or $(-\infty,-1]$ forces $\Re\arcsin w\to\pm\frac{\pi}{2}$ and $|\Im\arcsin w|\to\infty$. The real axis segment $(-1,1)$ maps to the real segment $(-\frac{\pi}{2},\frac{\pi}{2})$.

---

2. $z=\arccos w$

Principal formulae.

$$\boxed{\arccos w=\frac{\pi}{2}-\arcsin w}$$

$$\boxed{\arccos w=\frac{1}{i}\ \log\Big(w+\sqrt{w-1}\,\sqrt{w+1}\Big)}$$

Branch points & cuts.
Same as arcsin: branch points ±1; cuts $(-\infty,-1]\cup[1,\infty)$.

Target strip (canonical range).

$$\boxed{\arccos:\ \mathbb{C}\setminus\big((-\infty,-1]\cup[1,\infty)\big)\ \longrightarrow\ \{\,0<\Re z<\pi\,\}}$$

For $w\in(-1,1)$, $\arccos w\in(0,\pi)\subset\mathbb{R}$. For $w>1$, $\arccos w=i\ \operatorname{arcosh} w$ (so $\Re=0$); for $w<-1$, $\arccos w=\pi-i\ \operatorname{arcosh}(-w)$ (so $\Re=\pi$).

Boundary correspondence.
The two slits map to the vertical strip boundaries $\Re z=0$ and $\Re z=\pi$. The central interval $(-1,1)$ maps to the real segment $(0,\pi)$.

---

3. $z=\arctan w$

Principal formula.

$$\boxed{\arctan w=\frac{1}{2i}\ \log\left(\frac{1+i\,w}{1-i\,w}\right)}$$

Branch points & cuts.
Branch points at $w = \pm i$; principal cuts $i(-\infty, -1] \cup i[1, \infty)$ along the imaginary axis beyond $\pm i$.

Target strip (canonical range).

$$\boxed{\arctan: \ \mathbb{C} \setminus i\big((-\infty, -1] \cup [1, \infty)\big) \ \longrightarrow \ \{\, |\Re z| < \tfrac{\pi}{2} \,\}}$$

On $\mathbb{R}$, $\arctan \mathbb{R} = (-\frac{\pi}{2}, \frac{\pi}{2})$. Approaching $w = \pm i\, t$ $(t \geq 1)$ drives $\Re \arctan w \to \pm\frac{\pi}{2}$ and $|\Im| \to \infty$.

Symmetries.
$\arctan(-w) = -\arctan w$; conjugation across the real axis preserves values away from cuts.

---

4. $z = \operatorname{arccot} w$

Principal formulae.

$$\boxed{\operatorname{arccot} w = \frac{1}{2i} \log\left(\frac{w+i}{w-i}\right)}$$

$$\boxed{\operatorname{arccot} w = \frac{\pi}{2} - \arctan w}$$

Branch points & cuts.
As for arctan: branch points at $w = \pm i$; cuts $i(-\infty, -1] \cup i[1, \infty)$.

Target strip (canonical range).

$$\boxed{\operatorname{arccot}: \ \mathbb{C} \setminus i\big((-\infty, -1] \cup [1, \infty)\big) \ \longrightarrow \ \{\, 0 < \Re z < \pi \,\}}$$

On $\mathbb{R}$, $\operatorname{arccot} \mathbb{R} = (0, \pi)$. Approaching the cuts pins the image to $\Re z = 0$ or $\Re z = \pi$ with $|\Im| \to \infty$.

---

5. Relations and identities (principal branches)

$$\arcsin w + \arccos w = \frac{\pi}{2}, \qquad \arctan w + \operatorname{arccot} w = \frac{\pi}{2}.$$

$$\arcsin w = -\, i \ \operatorname{arsinh}(iw), \qquad \arccos w = \operatorname{arcosh} w \cdot \frac{1}{i} \ \ (\text{on } [1, \infty)).$$

$$\arctan w = \frac{1}{2i}\big(\log(1+iw) - \log(1-iw)\big), \quad \operatorname{arccot} w = \frac{1}{2i}\big(\log(w+i) - \log(w-i)\big).$$

6. Boundary/axis behavior (summary)

- arcsin: real on $(-1,1)$; the cuts map to $\Re z = \pm\frac{\pi}{2}$.
- arccos: real on $(-1,1)$; the cuts map to $\Re z = 0,\ \pi$.
- arctan: real on $\mathbb{R}$; the imaginary-axis cuts beyond $\pm i$ map to $\Re z = \pm\frac{\pi}{2}$.
- arccot: real on $\mathbb{R}$; the same cuts map to $\Re z = 0,\ \pi$.

Conformal maps: an overview catalogue (no details)

1. **Cayley (disk $\leftrightarrow$ right half-plane).**

$$\boxed{w = \frac{1+z}{1-z}} \qquad \text{inverse:} \quad z = \frac{w-1}{w+1}.$$

2. **Half-plane $\to$ half-plane (three-point normalization).**

$$\boxed{w = \frac{(z-a)(c-b)}{(z-b)(c-a)}}.$$

3. **Disk automorphisms.**

$$\boxed{\phi_{a,\theta}(z) = e^{i\theta}\frac{z-a}{1-\bar{a}z}}, \qquad |a|<1.$$

4. **Strip $|\Im z| < h/2 \to$ right half-plane.**

$$\boxed{w = \exp\Big(\frac{\pi z}{h}\Big)}.$$

5. **Strip $|\Im z| < h/2 \to$ unit disk.**

$$\boxed{\zeta = \frac{e^{\pi z/h}-1}{e^{\pi z/h}+1}}.$$

6. **Strip $\to$ strip (height scaling).**

$$\boxed{z \mapsto \frac{h_2}{h_1}\,z}.$$

7. **Band $a < \Im z < b \to$ centered strip.**

$$\boxed{z \mapsto \frac{\pi}{b-a}\Big(z - i\frac{a+b}{2}\Big)}.$$

8. **Sector $\{0 < \arg z < \alpha\} \to$ upper half-plane.**

$$\boxed{w = z^{\pi/\alpha}} \quad \text{(principal branch).}$$

9. **Wedge $\to$ disk.**
   Compose item (8) with Cayley (item 1).
10. **Upper half-plane $\to$ disk (prescribed triple).**

$$\boxed{\Phi(z) = \frac{(z-a)}{(z-\bar{a})} \cdot \frac{(b-\bar{a})}{(b-a)}}.$$

11. **Annulus $r < |z| < 1 \leftrightarrow$ strip.**

$$\boxed{w = \log z} \quad \text{maps to} \quad \log r < \Re w < 0.$$

12. **Annulus $\to$ disk.**
    Use $w = \log z$ then item (5).
13. **Half-plane minus disk $\to$ annulus.**

$$\boxed{\zeta = \frac{z-\alpha}{z+\alpha}}, \qquad \alpha^2 = a^2 - b^2.$$

14. **Disk with off-center hole $\to$ annulus.**
    Center via ϕ_a (item 3), then items (11)–(12).
15. **Punctured plane $\leftrightarrow$ cylinder.**

$$\boxed{w = \log z} \quad \text{(period } 2\pi i).$$

16. **Strip $\to$ punctured plane.**

$$\boxed{z \mapsto e^z}.$$

17. **Ray-slit plane $\mathbb{C} \setminus [0, \infty) \to$ half-plane.**

$$\boxed{w = \sqrt{z}} \quad \text{(choose branch)}.$$

18. **Segment-slit plane $\mathbb{C} \setminus [a, b] \to$ half-plane.**

$$\boxed{w = \sqrt{\frac{z-a}{z-b}}}.$$

19. **Exterior of $[-1, 1] \leftrightarrow$ exterior unit circle (Joukowski).**

$$\boxed{w = z + \frac{1}{z}}, \qquad z = \frac{w \pm \sqrt{w^2-4}}{2}.$$

20. **Exterior of ellipse $\leftrightarrow$ exterior unit circle.**

$$\boxed{w = \tfrac{1}{2}\left(cz + \frac{1}{cz}\right)} \quad (c \text{ from axes}).$$

21. **Right half-plane $\to$ vertical strip.**

$$\boxed{z \mapsto \frac{1}{\pi} \log \frac{z-1}{z+1}}.$$

22. **Upper half-plane → rectangle (SC/elliptic).**

$$\boxed{\zeta = \int^{z} \frac{dt}{\sqrt{(t-a)(t-b)(t-c)(t-d)}}}.$$

23. **Rectangle → disk (Jacobi sn).**

$$\boxed{\zeta = \operatorname{sn}\!\left(\frac{K(k)}{L}\,z;\ k\right)}.$$

24. **Half-disk ↔ half-plane.**
    Restriction of Cayley (item 1) and inverse.

25. **Lens (intersection of two disks) → disk.**
    Möbius normalizes the circles; then a power map; then Cayley.

26. **Half-strip $\{x > 0,\ 0 < \Im z < h\}$ → half-disk.**

$$\boxed{\exp(\pi z/h)} \quad \text{then power, then Cayley.}$$

27. **Polygon → half-plane (Schwarz–Christoffel).**

$$\boxed{\Phi'(z) = C \prod_k (z - z_k)^{\alpha_k - 1}}, \qquad \sum \alpha_k = 2.$$

28. **Upper half-plane → slit half-plane.**

$$\boxed{w = z + \frac{1}{z}} \quad \text{then a real Möbius adjustment.}$$

29. **Disk → disk with boundary arc removed.**
    Automorphism to put endpoints at ± 1, then boundary power z^{λ}.

30. **Finite Blaschke product (disk → disk, degree n).**

$$\boxed{B(z) = e^{i\theta} \prod_{k=1}^{n} \frac{z - a_k}{1 - \bar{a}_k z}}.$$

1. Cayley transform: $\mathbb{D} \leftrightarrow$ right half-plane

Map and inverse.

$$w = \frac{1+z}{1-z} \quad \text{maps} \quad \mathbb{D} = \{|z| < 1\} \text{ onto } \{\,\Re w > 0\,\}, \qquad z = \frac{w-1}{w+1}$$

is the inverse.

Why $\Re w > 0$.
Compute
$$\Re\left(\frac{1+z}{1-z}\right) = \frac{1-|z|^2}{|1-z|^2}.$$
Thus $\Re w > 0$ for $|z| < 1$, and $\Re w = 0$ for $|z| = 1$ (except $z = 1$, which maps to $w = \infty$). Therefore
$$|z| < 1 \iff \Re w > 0, \qquad |z| = 1 \iff \Re w = 0.$$

Boundary correspondence.
$z = 0 \mapsto w = 1, \quad z \to 1^- \mapsto w \to +\infty, \quad$ and $\partial\mathbb{D} \setminus \{1\}$ maps to the line $\Re w = 0$.
Circles/lines orthogonal to $\partial\mathbb{D}$ map to *vertical* lines in the w–plane.

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2. Three-point Möbius normalization: half-plane $\leftrightarrow$ half-plane

Cross-ratio map to the real axis.
Let L be a line or circle, and choose three distinct points $a, b, c \in L$. Define
$$\Phi(z) = \frac{(z-a)(c-b)}{(z-b)(c-a)}.$$
Then $\Phi(L) \subset \mathbb{R} \cup \{\infty\}$, with
$$\Phi(a) = 0, \qquad \Phi(b) = \infty, \qquad \Phi(c) = 1,$$
and the two sides of L map to the half-planes $\Im\Phi(z) \gtrless 0$.

Half-plane to half-plane via three boundary points.
Given half-planes H_1, H_2 with boundary arcs L_1, L_2, pick ordered boundary triples $a, b, c \in L_1$ and $A, B, C \in L_2$ (same cyclic order). Then
$$T(z) = \frac{(z-a)(c-b)}{(z-b)(c-a)} \cdot \frac{(C-A)}{(C-B)}$$
is a Möbius map sending $L_1 \to L_2$ with
$$a \mapsto A, \qquad b \mapsto B, \qquad c \mapsto C,$$
and taking the chosen side of H_1 onto the chosen side of H_2.

Specialization to the real line.
To send L to $\mathbb{R}$, use Φ above; then pre/postcompose with an *affine real* map
$$\xi \longmapsto \alpha\,\xi + \beta, \qquad \alpha > 0,$$
to obtain any target half-plane, e.g. $\Re w > 0$ or $\Im w > 0$.